Solving Problems and Designing Solutions: Thomas A. Edison

Chapter 7 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **How did Thomas Edison use his knowledge of energy transfer and transformation to solve problems?**

KEY IDEA

KEY IDEA **Engineering is iterative: PROBLEM → DESIGN → TEST → EVIDENCE → IMPROVE. Failure is information.**

EDISON'S LIGHT-BULB WORK

Source-supported facts. Read each, then write what you learned.

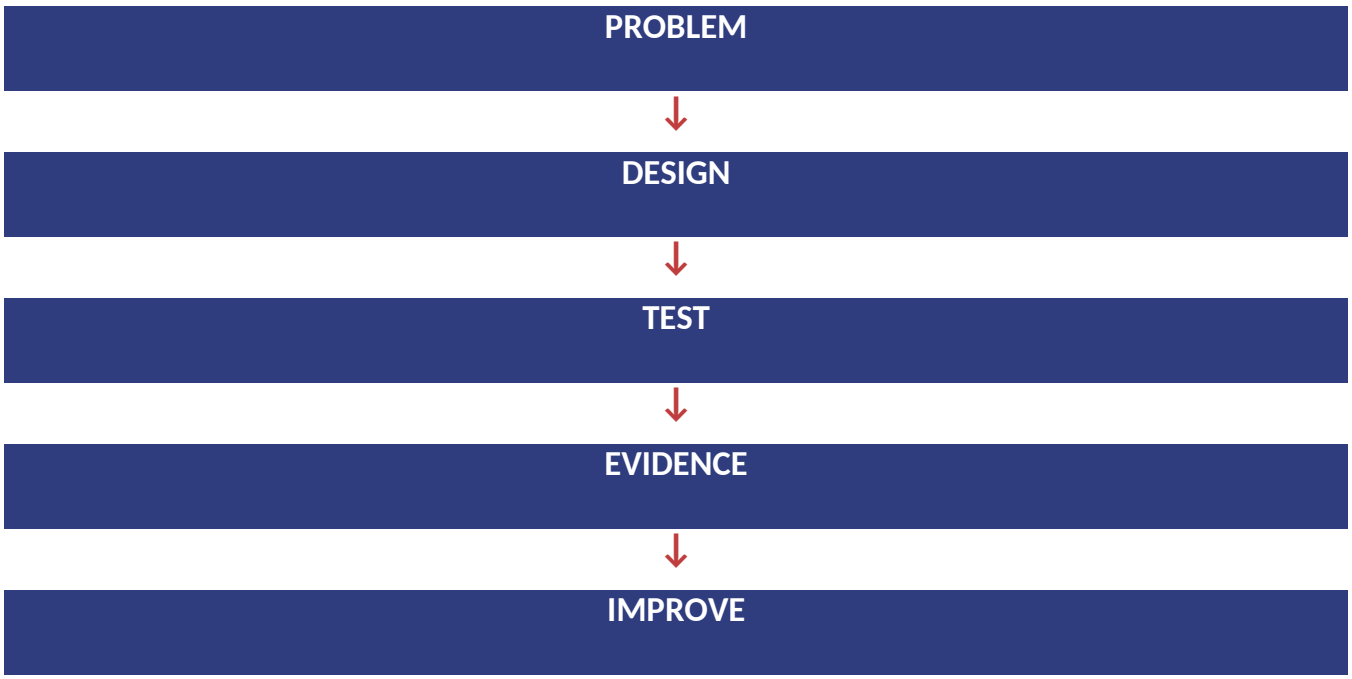
FACT	WHAT I LEARNED
Edison did NOT invent the first light bulb — he IMPROVED earlier designs.	
His team tested more than 6,000 materials for the filament.	
In 1879 he used a bamboo + carbon filament that lasted longer.	
Lewis Latimer improved the bulb too (wire-to-filament + stronger filament).	

The Engineering Cycle

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MODEL IT



FILL THE CYCLE

Trace the cycle with Edison's light-bulb work AND with your own device.

STEP	EDISON'S EXAMPLE	MY DEVICE
Problem	light bulbs did not last or were not practical	
Design		
Test		
Evidence		
Improve		

Unit Concept Map

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CONNECT THE WHOLE UNIT

Energy is the ability to cause change. Draw arrows to connect each idea to ENERGY.

change	**stored energy**	**motion**
collisions	**ENERGY**	**transformation**
speed	**engineering**	**transfer**

CHECK YOUR THINKING

Write one sentence that connects energy, change, and engineering using the whole unit.

CONNECT *Unit synthesis: energy causes change → motion & speed → transfer & collisions → transformation → engineering.*